(1) Laspeyre's Index

$$P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

$$= \frac{1185}{970} \times 100$$

$$= 122.16$$

(2)) Paasche's Index

$$P_{01}^{p} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$

$$= \frac{11254}{920} \times 100$$

$$= 122.28$$

(3) Fisher's Ideal Index

$$P_{01}^{F} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

$$= \sqrt{\frac{1185}{970} \times \frac{1125}{920}} \times 100$$

$$= \sqrt{1.2216 \times 1.2228} \times 100$$

$$= \sqrt{1.49377} \times 100$$

$$= 1.2222 \times 100$$

$$= 122.22$$

Example 6.8

Calculate the Dorbish and Bowley's price index number for the following data taking 2014 as base year.

Items	2014		2015	
	Prices (per kg)	Quantities (purchased)	Prices (per kg)	Quantities (purchased)
Oil	80	3	100	4
Pulses	35	2	45	3
Sugar	25	2	30	3
Rice	50	30	54	35
Cereals	35	2	40	3

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